Black Box Testing Report
Programming Language Ecosystem System

Modules Covered

- Main Menu
- Admin Login
- Developer Module
- Regular User Module
- Download System
- Rating and Feedback

1. EQUIVALENCE PARTITIONING (EP)

1.1 Main Menu

Test Case ID	Input	Partition Type	Expected Output
MM_EP_01	1–4 (valid options)	Valid	Navigate to module
MM_EP_02	0	Invalid	Show error
MM_EP_03	5+	Invalid	Show error
MM_EP_04	”abc”	Invalid	Show error

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 1
Enter Admin ID:
```

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 0
Invalid choice.
```

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 10
Invalid choice.
```

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: abc
Invalid input! Only numbers allowed.
```

1.2 Admin Login

Test Case ID	Input	Partition Type	Expected Output
AL EP 01	Correct username & password	Valid	Login successful
AL EP 02	Wrong username	Invalid	Login failed
AL EP 03	Wrong password	Invalid	Login failed
AL EP 04	Empty input	Invalid	Prompt error

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 1
Enter Admin ID: Aakash
Enter Password: 1234

--- Admin Menu ---
1. Perform Default Action
2. Approve Pending Language
3. Delete Language
4. View All Users
5. Manage Permissions
6. Check Feedback
7. Logout
Choice:
```

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 1
Enter Admin ID: hemanshu
Enter Password: 1234
Invalid credentials! Access Denied.
```

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 1
Enter Admin ID: Aakash
Enter Password: man
Invalid credentials! Access Denied.
```

```
=== Programming Language Ecosystem Interactive Menu ===
1. Login as Admin
2. Login as Developer
3. Login as Regular User
4. Exit
Enter choice: 1
Enter Admin ID:

1234
Enter Password:



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Invalid credentials! Access Denied.
```

1.3 Rating System

Test Case ID	Input	Partition Type	Expected Output
RT_EP_01	Rating 1–5	Valid	Accepted
RT_EP_02	Rating 0	Invalid	Reject
RT_EP_03	Rating 6+	Invalid	Reject

```
Enter rating (1-5): 1
[Regular User] Rated language 1 stars.
```

```
Choice: 1
Enter rating (1-5): 0
[Error] Rating out of range! Must be between 1 and 5. Got: 0
```

```
Choice: 1
Enter rating (1-5): 0
[Error] Rating out of range! Must be between 1 and 5. Got: 0
Enter rating (1-5): 6
[Error] Rating out of range! Must be between 1 and 5. Got: 6
Enter rating (1-5):
```

2. BOUNDARY VALUE ANALYSIS (BVA)

2.1 Main Menu

Test Case ID	Input	Boundary Type	Expected Output
MM_BVA_01	1	Lower boundary	Valid
MM_BVA_02	4	Upper boundary	Valid
MM_BVA_03	0	Below lower	Error
MM_BVA_04	5	Above upper	Error

2.2 Rating System

Test Case ID	Input	Boundary Type	Expected Output
RT_BVA_01	1	Min valid	Accept
RT_BVA_02	5	Max valid	Accept
RT_BVA_03	0	Below min	Reject
RT_BVA_04	6	Above max	Reject

3. DECISION TABLE TESTING

3.1 Admin Login

Conditions	Case1	Case2	Case3	Case4
Correct Username	Y	Y	N	N
Correct Password	Y	N	Y	N

Action	Case1	Case2	Case3	Case4
Login Success	X			
Error		X	X	X

3.2 Rating System

Conditions	Case1	Case2	Case3
First time rating	Y	N	Y
Valid rating (1–5)	Y	Y	N

Action	Case1	Case2	Case3
Accept	X		
Reject		X	X

4. STATE TRANSITION TESTING

Test Case ID	Initial State	Input	Next State
ST_01	QUEUED	Start	DOWNLOADING
ST_02	DOWNLOADING	Complete	COMPLETED
ST_03	DOWNLOADING	Pause	PAUSED
ST_04	PAUSED	Resume	DOWNLOADING
ST_05	DOWNLOADING	Error	FAILED

5. ERROR GUESSING

5.1 Main Menu

Test Case ID	Scenario	Expected Output
MM_EG_01	Non-integer input	Reject
MM_EG_02	Large number	Error
MM_EG_03	Empty input	Prompt

5.2 Admin Login

Test Case ID	Scenario	Expected Output
AL_EG_01	Very long input	Handle safely
AL_EG_02	Case sensitivity issue	Validate properly
AL_EG_03	Special characters	Reject

5.3 Rating System

Test Case ID	Scenario	Expected Output
RT_EG_01	Multiple ratings	Reject
RT_EG_02	Negative rating	Reject
RT_EG_03	No rating but feedback	Error